

Chapter 3: The Cox Regression Model

Modelling Survival Data in Medical Research

Collett (2023), 4th ed.

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Part 2

Chapter 3 Outline

1. Comparing Models: $-2 \log \hat{L}$
2. Model Selection Strategy
3. The Lasso
4. Non-Linear Terms

The $-2 \log \hat{L}$ Statistic — Intuition

After fitting a model, we have the **maximised partial likelihood** $\hat{L}$.

Why $-2 \log \hat{L}$?

$\hat{L}$ is a product of small probabilities, so $0 < \hat{L} < 1$.

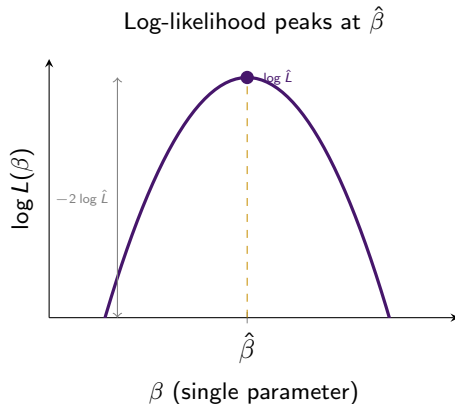
$\log \hat{L} < 0$ always.

$-2 \log \hat{L} > 0$ always.

Multiplying by -2 gives the statistic the right chi-squared distribution.

For a given dataset

Smaller $-2 \log \hat{L} \Rightarrow$ better fit. Not comparable across datasets; only useful comparing models on the *same* data.



The Likelihood Ratio Test for Nested Models

Setup: Model 1 has p variables; Model 2 adds q more ($p + q$ total).

Likelihood ratio test statistic

$$\Delta = (-2 \log \hat{L}_1) - (-2 \log \hat{L}_2) \geq 0$$

Under $H_0 : \beta_{p+1} = \dots = \beta_{p+q} = 0$: $\Delta \sim \chi_q^2$

Logic: adding variables always reduces $-2 \log \hat{L}$.
The question is whether the reduction Δ is *significantly large* for q extra parameters.

Decision:

- ▶ $P < 0.05$: extra variables significantly improve the model $\Rightarrow$ keep them
- ▶ $P > 0.05$: simpler model adequate $\Rightarrow$ keep Model 1

The LRT is valid because standard likelihood theory applies to the partial likelihood.

Example 3.3 — Breast Cancer: Likelihood Ratio Test

Is HPA staining a significant predictor of survival?

Model	Variables	$-2 \log \hat{L}$
Null	—	173.968
With X (HPA staining)	X	170.096
$\Delta = 3.872$ on 1 d.f.		$P = 0.049$

$P = 0.049$: moderate evidence HPA staining affects survival.

$HR = e^{0.908} = 2.48$: positive staining $\Rightarrow$ 2.48 $\times$ the hazard.

Connection to Chapter 2:

Log-rank: $W_L = 3.515$, $P = 0.061$.

LRT: $\Delta = 3.872$, $P = 0.049$.

Similar conclusions — the tie at $t = 26$ causes the slight numerical difference.

AIC and BIC: When Models Are Not Nested

Sometimes we want to compare models where neither contains the other as a special case.

Akaike Information Criterion (AIC)

$$\text{AIC} = -2 \log \hat{L} + 2q$$

q = number of β -parameters. The $+2q$ **penalises complexity**: adding useless variables increases AIC.

Smaller AIC = better model.

Bayesian Information Criterion (BIC)

$$\text{BIC} = -2 \log \hat{L} + q \log(d)$$

d = number of *uncensored* events.

Heavier penalty than AIC for large samples.
Tends to favour simpler models.

Smaller BIC = better model.

- ▶ Use AIC/BIC when comparing many non-nested model alternatives simultaneously
- ▶ AIC and BIC usually agree; when they differ, BIC is more conservative

AIC/BIC can also be used as an alternative to hypothesis testing when comparing nested models.

Example 3.4 — Hypernephroma: Five Models Compared

36 kidney cancer patients. Factors: age group (3 levels), nephrectomy (yes/no).

Model	Terms	$-2 \log \hat{L}$	d.f.
(1) Null	—	177.667	0
(2) Age group only	α_j	172.172	2
(3) Nephrectomy only	ν_k	170.247	1
(4) Both main effects	$\alpha_j + \nu_k$	165.508	3
(5) Both + interaction	$\alpha_j + \nu_k + (\alpha\nu)_{jk}$	162.479	5

(4) vs. (5): $\Delta = 3.029$, $P = 0.220$ — no interaction.

(3) vs. (4): $\Delta = 4.739$, $P = 0.094$ — age adds.

(2) vs. (4): $\Delta = 6.664$, $P = 0.010$ — nephrectomy needed.

Conclusion: Model (4) is best.

The Challenge: Which Variables to Include?

In most studies, many potential explanatory variables are measured. Including *all* of them causes problems:

- ▶ **Overfitting:** model fits the noise in the sample, not the signal
- ▶ **Wide standard errors:** estimated coefficients are unstable
- ▶ **Unreliable predictions:** the model does not generalise to new patients

Goals of model selection:

- ▶ Smallest set of variables that explains the data
- ▶ Avoid including unnecessary variables
- ▶ Avoid excluding important variables
- ▶ Clinically interpretable result

In practice there are usually *several* equally good models — not one unique “best” model. Report this uncertainty honestly.

Rule of thumb

At least **10–20 events per variable**. With 48 patients and 30 deaths, fitting more than 2–3 variables is risky.

The Recommended 4-Step Strategy

- 1 **Screen individually.** Fit each variable on its own. Compare $-2 \log \hat{L}$ to the null model. Identify those with $P < 0.15$ (a generous threshold to avoid missing important variables).
- 2 **Fit all promising variables together.** Then, in the full model, try omitting each variable one at a time. If omitting a variable causes only a small increase in $-2 \log \hat{L}$ ($P > 0.15$), drop it. Repeat until no more variables can be dropped.
- 3 **Check what was excluded.** Variables that were unimportant alone may become important in combination. Add them one at a time to the model from Step 2 and retain any that are significant.
- 4 **Final check.** Confirm no term in the model can be removed, and no omitted term can be added, without a significant change in $-2 \log \hat{L}$.

Use 15% as your threshold, not 5%. Being too strict at selection risks omitting important variables.

Example 3.5 — Multiple Myeloma: Full Selection

Step 1 — individual screening

Null model: $-2 \log \hat{L} = 215.940$

Variable	$-2 \log \hat{L}$	P
Bun	207.453	0.004**
Hb	211.068	0.027*
Protein	213.890	0.152
Ca	215.494	0.313
Age	215.817	0.568
Sex	215.906	0.664
Pcells	215.875	0.637

Keep Bun, Hb, Protein for Step 2.

Step 2 — fit Bun+Hb+Protein together

$-2 \log \hat{L} = 200.503$

Omit	Δ	P
Bun	+9.33	0.002**
Hb	+3.14	0.076*
Protein	+2.44	0.119

Protein: $P = 0.119 < 0.15$, borderline. Drop it.

Step 3 — add others to Bun+Hb:

Age, Sex, Ca, Pcells: each $\Delta < 0.5$, all $P > 0.5$.

Final model: Bun + Hb only.

Automatic Variable Selection: Pros and Cons

Forward selection

Start with null. Add variables one at a time, choosing the one giving largest Δ .

Stop when no more variables are significant.

Backward elimination

Start with all variables. Remove one at a time, dropping the one giving smallest Δ when removed.

Stop when all remaining are significant.

Stepwise

Like forward, but after each addition, check whether any earlier variable can now be removed.

Problems with automatic procedures:

- ▶ Each gives *one* model, not a set of equally good alternatives
- ▶ Results depend on the stopping rule chosen
- ▶ May violate the hierarchic principle
- ▶ Lead to overfitting when p is large relative to n
- ▶ P -values from the selected model are *not* valid (selection inflation)

Use automatic procedures as a starting point only.
The 4-step manual strategy provides much better insight.

Model Selection for Treatment Studies

When the primary goal is to estimate a **treatment effect**, the strategy differs slightly.

- 1 **Select prognostic variables first**, ignoring treatment. Use the 4-step strategy on all other variables.
- 2 **Add the treatment variable** to the model from Step 1. The treatment effect is then adjusted for all other prognostic variables.
- 3 **Check for interactions** between treatment and other variables (e.g. treatment $\times$ age). If these are significant, the treatment effect differs by covariate level.

Why this order matters (Example 3.12)

In the prostatic cancer study, the unadjusted treatment HR is 0.14. After adjusting for tumour size and Gleason index, the adjusted HR is 0.33. Without adjustment, the treatment benefit is overestimated by a factor of ~ 2.4 . This is confounding.

The Lasso: Variable Selection via Shrinkage

Maximise the partial likelihood with a **penalty** on coefficient size:

$$L_{\lambda}(\beta) = L(\beta) - \lambda \sum_{j=1}^p |\beta_j|$$

$\lambda \geq 0$ = lasso parameter. Larger $\lambda \Rightarrow$ more shrinkage.

Key property: the lasso sets some $\hat{\beta}_j$ to *exactly* zero. Variables with $\hat{\beta}_j = 0$ are excluded; those with $\hat{\beta}_j \neq 0$ are included.

When is the lasso useful?

- ▶ Many potential variables ($p > 20$)
- ▶ Some variables are highly correlated
- ▶ Traditional step-by-step selection is too slow
- ▶ Prediction is more important than inference

Before applying: standardise all variables (zero mean, unit variance) so the penalty is applied equally.

Optimal λ : chosen by cross-validated partial log-likelihood.

The Lasso Trace and Optimal λ

Lasso trace: plot $\hat{\beta}_j$ vs. λ for each variable j .

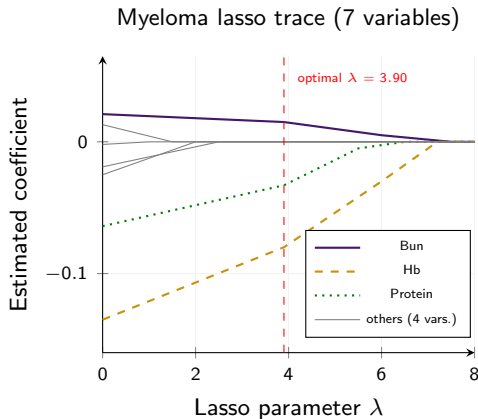
- ▶ At $\lambda = 0$: standard Cox estimates
- ▶ As λ increases, all $|\hat{\beta}_j|$ decrease (shrinkage)
- ▶ Eventually each $\hat{\beta}_j$ hits zero; the variable is excluded
- ▶ The trace is **piecewise linear**

Choosing λ :

Use **cross-validated partial log-likelihood**.

Split data into k folds. For each λ , fit the model on $k - 1$ folds, evaluate on the remaining fold.

Choose λ that maximises the average cross-validated log-likelihood.



The Lasso: What It Selects and Why

Example 3.7 — Multiple Myeloma

At optimal $\lambda = 3.90$, three variables have non-zero coefficients: **Bun, Hb, Protein**.

Variable	Lasso estimate	Cox estimate	Shrinkage?
Bun	0.015	0.020	Yes (smaller)
Hb	-0.080	-0.110	Yes (closer to 0)
Protein	-0.330	-0.617	Yes (closer to 0)
Age, Sex, Ca, Pcells	0	≈ 0	Set to zero

Lasso agrees with the 4-step manual selection (Bun + Hb, with Protein borderline) and provides shrunk coefficients.

Caution: lasso estimates are **biased** toward zero. Use lasso to **select variables**, then refit a standard Cox model with those variables.

Is the Linearity Assumption Always Reasonable?

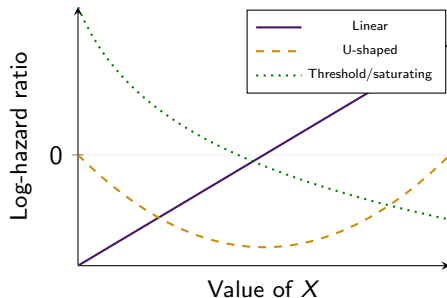
When we include X as βX , we assume the **log-hazard ratio is linear** in X .

When linearity may fail:

- ▶ Very low *and* very high values of X both increase risk (U-shaped)
- ▶ The effect of X plateaus at extreme values
- ▶ The relationship has a threshold (no effect below a critical value)

Example: serum haemoglobin (Hb).

Both anaemia (low Hb) and certain high values may be associated with worse prognosis. A linear assumption may miss this.



How to Test for Non-Linearity

Procedure (using the factor method)

- 1 Group X into 4–5 categories $\Rightarrow$ factor A
- 2 Fit A as a **factor** ($m - 1$ parameters): $-2 \log \hat{L}_A$
- 3 Fit X as a **linear variate** (1 parameter): $-2 \log \hat{L}_X$
- 4 $\Delta = (-2 \log \hat{L}_X) - (-2 \log \hat{L}_A) \sim \chi^2_{m-2}$
- 5 Significant $\Delta \Rightarrow$ linearity does not hold

Example 3.8 — Hb in the myeloma model

Group Hb into 4 levels (quartiles): ≤ 7 , 7–10, 10–13, > 13 .

Factor model: $-2 \log \hat{L}_A = 200.417$.

Linear model: $-2 \log \hat{L}_X = 203.891$.

$\Delta = 3.474$ on 2 d.f., $P = 0.176$ — **not significant**.

$\Rightarrow$ Linear term in Hb is adequate.

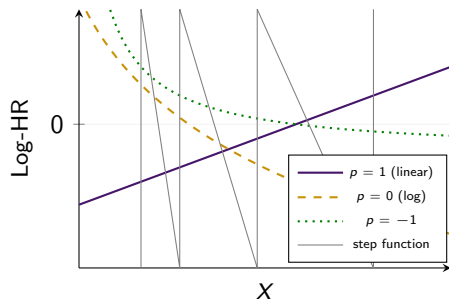
Fractional Polynomials: Smooth Non-Linear Modelling

A fractional polynomial of order m in X :

$$\beta_1 X^{p_1} + \dots + \beta_m X^{p_m}, \quad p_j \in \{-2, -1, -0.5, 0, 0.5, 1, 2, 3\}, \quad X^0 \equiv \log X$$

Advantages over alternatives:

- ▶ *vs. polynomials*: FP don't oscillate wildly at tails; flexible with just 1–2 terms
- ▶ *vs. step functions*: no arbitrary cutpoints; no information loss; biologically natural
- ▶ *Practical*: $m = 1$ tries 8 powers; $m = 2$ tries 28 pairs. Pick the best-fitting.



Questions?

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